

ECO387C - Fall 2024

Problem Set 11

Raymond He & Husang Kim

You may work in groups, but please acknowledge any contributions from others and turn in separate assignments.

Question 1 - Solow model without technological change

1. Recall that

$$R(t) = Af'(k(t))$$

Suppose that $0 < k(0) < k^*$. Since $f''(\cdot) < 0$ by the neoclassical assumptions, as $k(t)$ rises monotonically over time, f' falls over time and thus $R(t)$ falls monotonically over time. Similarly,

$$w(t) = Af(k(t)) - Af'(k(t))k(t)$$

Differentiating the right-hand side with respect to $k(t)$, we get

$$f'(k(t)) - f'(k(t)) - k(t)f''(k(t)) = -k(t)f''(k(t)) > 0$$

Hence $w(t)$ moves positively with $k(t)$, and since $k(t)$ is monotonically rising over time, this means that the wage rate $w(t)$ is also monotonically rising over time. Thus, we have shown that both $w(t)$ and $R(t)$ are monotone functions of time.

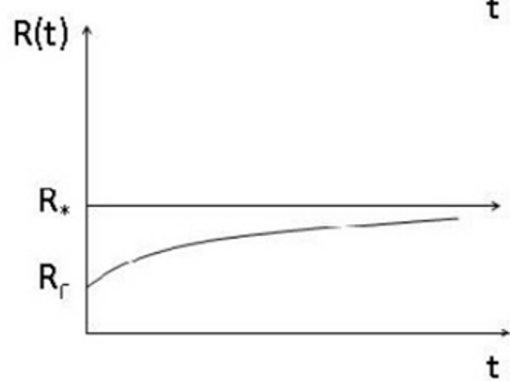
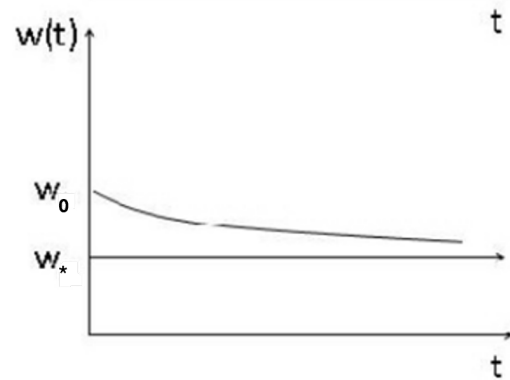
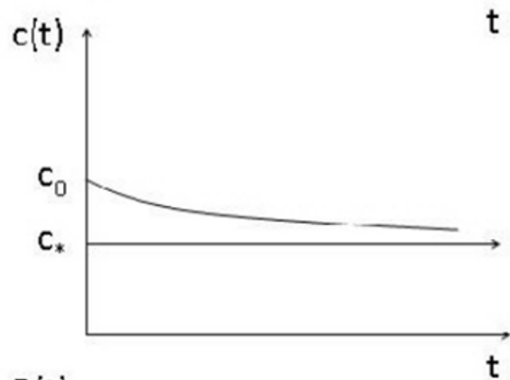
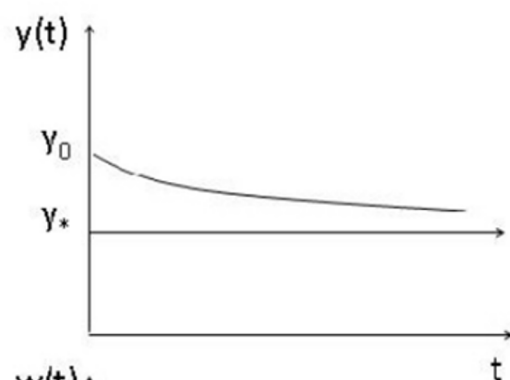
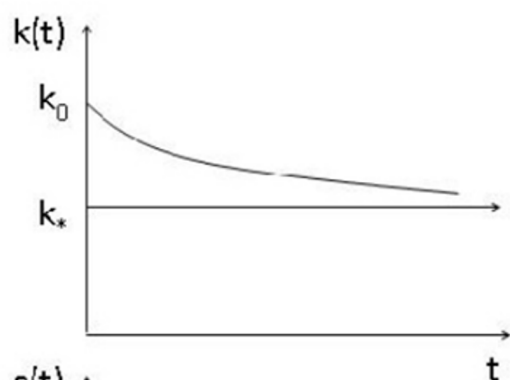
2. From (a) we know that $k(t)$ falls monotonically if $k(0) > k^*$ and $k(t)$ rises monotonically if $0 < k(0) < k^*$. First suppose $k(0) > k^*$. The time paths for $k(t)$, $y(t)$, $c(t)$, $w(t)$, and $R(t)$ are given by the following graph.

If $k(0) > k^*$, then $k(t)$ asymptotes downwards towards k^* . All we know about this time path is that it will be monotonically falling; the shape of the time path depends on the value of $k(0)$.

Since $y(t) = f(k(t))$ and $f'(\cdot) > 0$, because $k(t)$ is monotonically falling, this means that $y(t)$ is monotonically falling.

$c(t) = (1 - s)f(k(t))$ implies that if $k(t)$ is monotonically falling, then $c(t)$ is monotonically falling because $f'(\cdot) > 0$.

We showed in (a) that $w(t)$ moves positively with $k(t)$. Since $k(t)$ falls monotonically, we know that $w(t)$ falls monotonically.



Finally, we showed in (a) that $R(t)$ moves negatively with $k(t)$. Since $k(t)$ falls monotonically, we know that $R(t)$ rises monotonically. So, this time path will start at some $R(0)$ and then asymptote upwards to some $R^* = f'(k^*)$, with $R^* > R(0)$.

When $0 < k(0) < k^*$, $k(t)$ rises monotonically, $y(t)$ rises monotonically, $c(t)$ rises monotonically, $w(t)$ rises monotonically, and $R(t)$ falls monotonically.

3.

$$F(K, L, A) = AK^\alpha L^{1-\alpha} \Rightarrow \dot{k}(t) = sAk(t)^\alpha - (n+d)k(t)$$

Apply the change of variable ($x(t) = k(t)^{1-\alpha}$) and differentiate w.r.t. time:

$$\dot{x}(t) = (1-\alpha)k(t)^{-\alpha}\dot{k}(t) = (1-\alpha)k(t)^{-\alpha} \left[sAk(t)^\alpha - (n+d)k(t) \right]$$

This implies

$$\dot{x}(t) = (1-\alpha)sA - (1-\alpha)(n+d)x(t)$$

This is an ordinary linear differential equation in $x(t)$. First, we find the particular solution:

$$\dot{x}(t) = 0 \Rightarrow (1-\alpha)sA - (1-\alpha)(n+d)x^* = 0 \Rightarrow x^* = \frac{sA}{n+d}.$$

Hence, the entire family of solutions is:

$$x(t) = \frac{sA}{n+d} + c_1 \exp \left[-(1-\alpha)(n+d)t \right].$$

Evaluating at $t = 0$ and using our initial condition $x(0) = x_0 = k_0^{1-\alpha}$:

$$x_0 = \frac{sA}{n+d} + c_1 \Rightarrow c_1 = x_0 - \frac{sA}{n+d}.$$

Hence, the solution to this differential equation:

$$x(t) = \frac{sA}{n+d} + \left(x_0 - \frac{sA}{n+d} \right) \exp \left[-(1-\alpha)(n+d)t \right].$$

Since $x(t) = k(t)^{1-\alpha}$

$$k(t) = \left[\frac{sA}{n+d} + \left(x_0 - \frac{sA}{n+d} \right) \exp \left[-(1-\alpha)(n+d)t \right] \right]^{\frac{1}{1-\alpha}}.$$

The solution shows that starting from $k(0)$, $k(t)$ goes to $k^* = \left[\frac{sA}{n+d} \right]^{\frac{1}{1-\alpha}}$ as $t \rightarrow \infty$ in a monotonic fashion. This is consistent with the sketch in the previous question. The gap between $k(0)$ and k^* narrows at the exponential rate $(1-\alpha)(n+d)$.

Question 2 - Solow model with technological change

Recall the result from the class:

$$\dot{k}_t = sf(k_t) - (g_A + n + \delta)k_t$$

where $f(k) = k^\alpha$. The steady state value of capital k^* where $\dot{k}_t = 0$ satisfies:

$$\frac{f(k^*)}{k^*} = \frac{g_A + n + \delta}{s} \Leftrightarrow k^* = \left(\frac{s}{g_A + n + \delta} \right)^{\frac{1}{1-\alpha}}$$

Any initial value of $k_0 > 0$ will converge monotonically to k^* . Note that other endogenous variables can be expressed in terms of k_t :

$$w_t = A_t f(k_t) - A_t f'(k_t)k_t = A_t(1 - \alpha)k_t^\alpha$$

$$R_t = f'(k_t) = \alpha k_t^{\alpha-1}$$

$$K_t = k_t A_t L_t$$

$$Y_t = f(k_t)A_t L_t = k_t^\alpha A_t L_t$$

These relationships suggest that w_t , R_t and Y_t are positively correlated with k_t , whereas R_t is negatively correlated.

1. A permanent increase in s will define a new steady state level of capital (k) greater than the old one.
 - Since the new steady-state k is higher, the balanced growth path of wage shifts upward. The log wage grows with the growth rate g_A .
 - The new steady-state interest rate is lower than the old one. The interest rate converges to the new steady state.
 - Since the new steady-state k is higher, the balanced growth path of K shifts upward. The log K grows linearly with the growth rate $g_A + n$.
 - Since the new steady-state k is higher, the balanced growth path of Y shifts upward as well. The log Y grows linearly with the growth rate $g_A + n$.
2. A permanent increase in n will define a new steady state level of capital (k) smaller than the old one. The capital and interest rate will converge to the new steady state. The log wage shifts to a new BGP with the same growth rate g_A . While $\log K$ and $\log Y$ also shift to a new BGP, the slope on the new growth path is greater than the old BGP.
3. A permanent increase in g_A will define a new steady state level of capital (k) smaller than the old one. The capital and interest rate will converge to the new steady state. The log wage shifts to a new BGP with the new growth rate larger than the old one. $\log K$ and $\log Y$ also shift to a new BGP, the slope on the new growth path is greater than the old BGP.

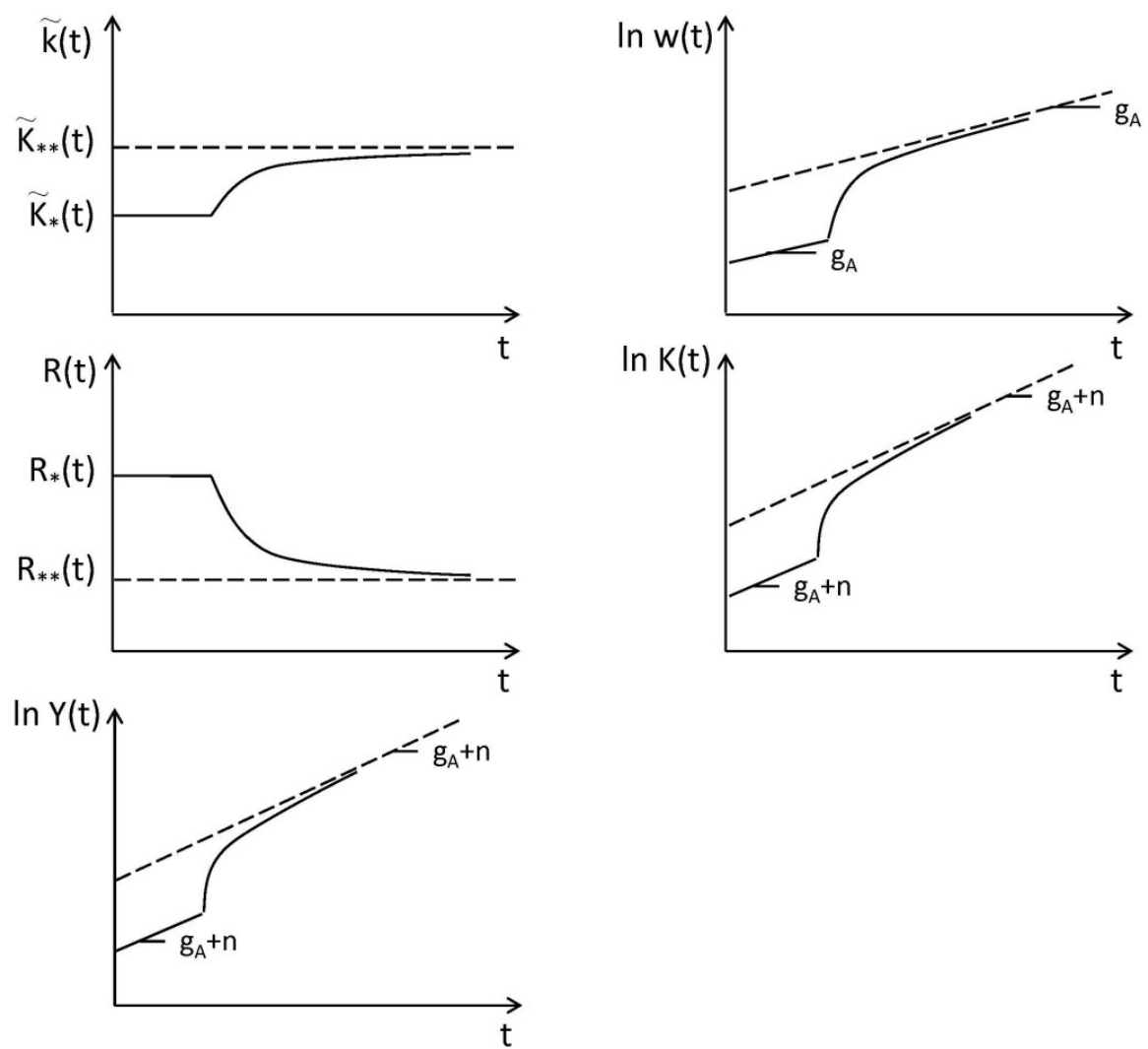


Figure 1: Q2-1

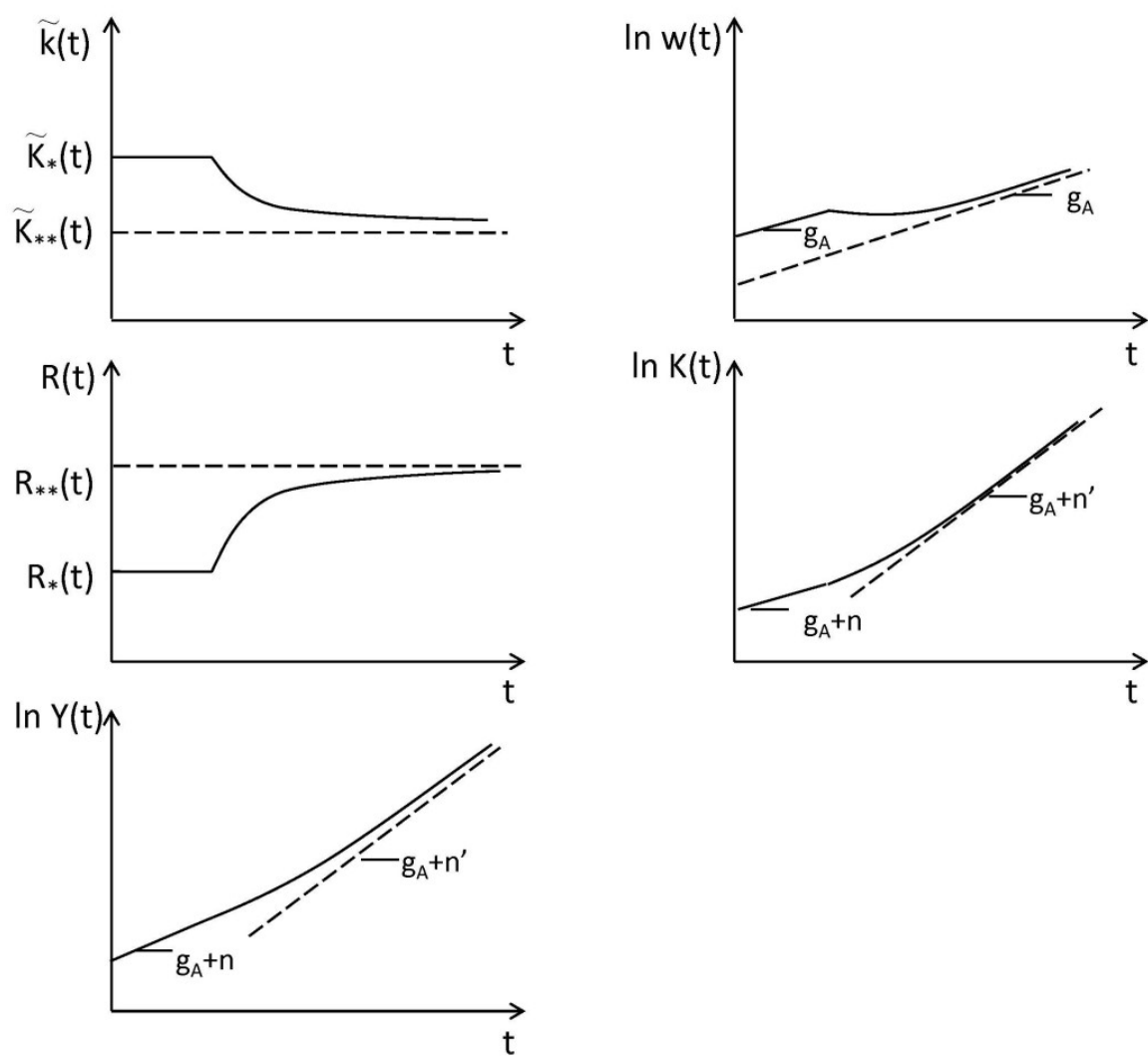


Figure 2: Q2-2

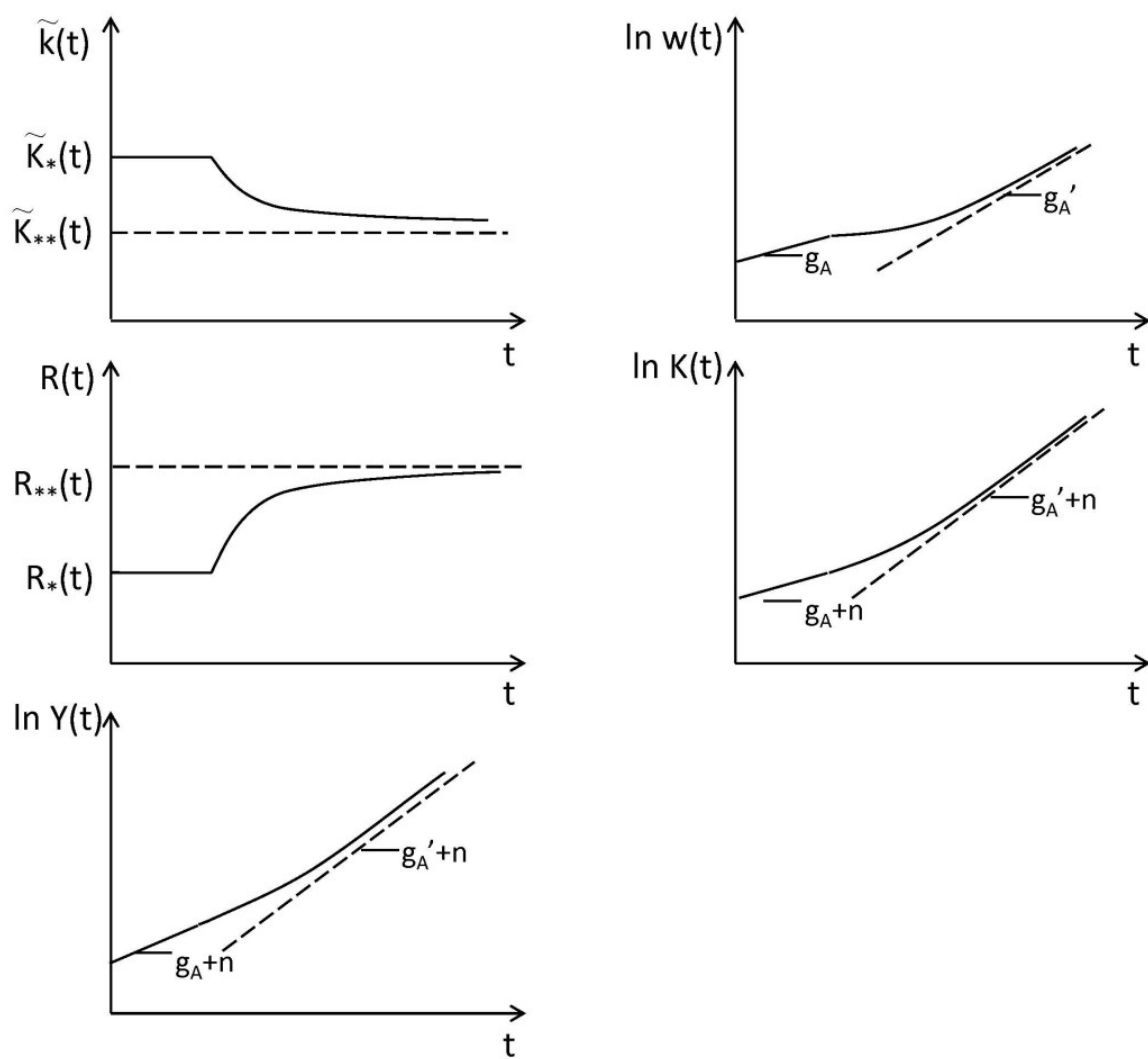


Figure 3: Q2-3